

Theory

Gaussian Beam Optics

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June 10, 2026

Abstract

The purpose of the document is to work through the theory and experimental design for optimizing second-harmonic generation (SHG) at 1064 nm using a periodically poled lithium niobate (PPLN) crystal ($50 \times 2 \times 0.5 \text{ mm}^3$). In this document, I define Gaussian beam parameters, including beam waist, Rayleigh range, divergence, and confocal parameter, in order to determine the optimal focusing geometry. The Boyd-Kleinman condition for maximum CW SHG efficiency ($L/b = 2.84$) is a target focused waist of approximately $54.6 \text{ }\mu\text{m}$, but experimental constraints redefines a target of $77\text{--}80 \text{ }\mu\text{m}$ ($L/b \approx 1.3 - 1.4$). Starting from a collimated 0.56 mm beam at 1064 nm with a measured divergence of 1.21 mRad , the collimator waist is estimated to be $\omega_{01} \approx 280 \text{ }\mu\text{m}$, located 6.2 mm behind the collimator face. That being said, the calculated waist radius of roughly 0.279 mm would result in a beam larger than the crystal itself can accommodate. To account for this, I used an ABCD ray-transfer matrix to propagate a beam, constructing two equations relating a chosen lens focal length f to the required lens position d_1 and image distance d_2 , with magnification $\alpha \approx 0.286$.

1 Experiment Specifics

1. Wavelength: 1064 nm
2. Collimator output beam size: $0.56 \text{ mm } \frac{1}{e^2}$ diameter
3. PPLN crystal: $50 \times 2 \times 0.5 \text{ mm}^3$
4. Previous best measured output: 57 mW green, PPLN at 59.5°C
5. Previous calculated waist: about $45 \text{ }\mu\text{m}$
6. Desired next target: about $77\text{--}80 \text{ }\mu\text{m}$

2 Definitions and Equations

Beam Diameter ($\frac{1}{e^2}$): The $\frac{1}{e^2}$ beam diameter captures the diameter where the intensity of the beam drops to 13.53%; in other words, 86.47% of the beam intensity lies within the given diameter.

Beam Waist (ω_0): The beam waist is the portion of the beam where the radius is minimized, denoted ω_0 . The tighter the focus the smaller the waist, but this introduces a large amount of divergence, as seen in the z -dependent beam width equation

$$\omega^2(z) = \omega_0^2 \left[1 + \left(\frac{\lambda z}{\pi \omega_0^2} \right)^2 \right]. \quad (1)$$

Small waists are achieved by a “short focal length and a high numerical aperture largely filled by the input beam.”[?]

Radius of Curvature (R_g): The radius of curvature is defined by Eq. (??). At its boundaries, $\lim_{z \rightarrow 0} R_g(z) = \infty$ where the wavefront is perfectly flat and acts like a section of a circle with infinite radius, and $\lim_{z \rightarrow \infty} R_g(z) = z$ where the wavefront averages out to a radius equal to the distance from the waist.

$$R_g(z) = z \left[1 + \left(\frac{\pi \omega_0^2}{\lambda z} \right)^2 \right]. \quad (2)$$

Rayleigh Range (z_0): The Rayleigh range is the distance at which the cross-sectional area of the beam is no more than twice that of the beam waist, ensuring at least 50% of the beam is retained.[?]

$$z_0 = \frac{\pi \omega_0^2}{\lambda}. \quad (3)$$

Beam Divergence (θ): The beam divergence is the slope of the linear component of beam expansion (the far-field expansion), measured as a half-angle dependent on the wavelength and inversely on the waist:

$$\theta = \frac{\lambda}{\pi \omega_0}. \quad (4)$$

Confocal Parameter (b): The confocal parameter uses the Rayleigh range to quantify the useful depth of focus—it is “the total length of the focused region around the beam waist.”[?]

Boyd–Kleinman Focusing: $h = 1.068$ at $L/b = 2.84$ [?]; $h = 1.066$ at $L/b = 2.81$ [?]. Here h is the conversion efficiency, L is the crystal length, and $b = 2z_0$ is the confocal parameter. We work backwards from these values to find the optimal waist; see Section ??.

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3 Calculations

3.1 Initial Beam Parameters

Starting from a 0.56 mm $1/e^2$ beam diameter at 1064 nm, what are the approximate Gaussian beam parameters?

Using Eq. (??):³

$$z_{01} = \frac{\pi \omega_{01}^2}{\lambda} = \frac{\pi \cdot (0.00028)^2}{1.064 \times 10^{-6}} \approx 0.23149 \text{ m} \approx 23.15 \text{ cm}.$$

This results in $L/b = 0.108$, far from the Boyd–Kleinman optimum.

Using Eq. (??):

$$\theta = \frac{\lambda}{\pi \omega_{01}} = \frac{1.064 \times 10^{-6}}{\pi \cdot 0.00028} \approx 1.210 \times 10^{-3} \text{ rad}.$$

3.2 Boyd–Kleinman Focusing Targets

3.2.1 Optimum: $L/b = 2.84$ [?]

Given $L/b = 2.84$ and $L = 50 \text{ mm}$.

$$b = \frac{L}{2.84} = \frac{0.05}{2.84} \text{ m}.$$

Using Eq. (??):

$$\frac{0.05}{2.84} = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot 2.84}} \approx 54.602 \mu\text{m}.$$

3.2.2 Optimum: $L/b = 2.81$ [?]

Given $L/b = 2.81$ and $L = 50 \text{ mm}$.

$$b = \frac{L}{2.81} = \frac{0.05}{2.81} \text{ m}.$$

Using Eq. (??):

$$\frac{0.05}{2.81} = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot 2.81}} \approx 54.892 \mu\text{m}.$$

3.2.3 Conservative rule of thumb: $L/b = 1$ [?]

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Given $L/b = 1$ and $L = 50 \text{ mm}$.

$$b = L = 0.05 \text{ m}.$$

Using Eq. (??):

$$0.05 = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi}} \approx 92.017 \mu\text{m}.$$

3.2.4 Working Backwards from Target Waists

To recover a desired ω_{02} , we solve for $\xi \equiv L/b$:

1. $\omega_{02} = 45 \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (45 \times 10^{-6})^2} \approx 4.181.$$

This ratio is too large (waist too tight), leading to excessive diffraction and reduced efficiency.

2. $\omega_{02} = 77 \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (77 \times 10^{-6})^2} \approx 1.428.$$

Within the proper range.

3. $\omega_{02} = 80 \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (80 \times 10^{-6})^2} \approx 1.323.$$

Within the proper range.

3.3 Collimator Waist Estimation

Repeating the analysis without assuming the beam waist is at the collimator face.⁵

1. Using Eq. (??) to recover the waist from the measured divergence:

$$\omega_{01} = \frac{\lambda}{\pi\theta} = \frac{1.064 \times 10^{-6}}{\pi \cdot 0.00121} \approx 279.9 \mu\text{m}.$$

2. Using the beam propagation equation (Eq. (??), from Eq. 2-27 in [?]):

$$\omega^2(z) = \omega_{01}^2 \left[1 + \left(\frac{\lambda z}{\pi \omega_{01}^2} \right)^2 \right]. \quad (5)$$

Solving for z :

$$\frac{\omega^2(z)}{\omega_{01}^2} - 1 = \left(\frac{\lambda z}{\pi \omega_{01}^2} \right)^2 \Rightarrow z = \pm \left(\sqrt{\frac{\omega^2(z)}{\omega_{01}^2} - 1} \right) \frac{\pi \omega_{01}^2}{\lambda}.$$

Setting $\omega(z) = 0.00028$ m (the $\frac{1}{e^2}$ radius at the collimator aperture):

$$z \approx \pm \left(\sqrt{\frac{(0.00028)^2}{(0.0002799)^2} - 1} \right) \frac{\pi \cdot (0.0002799)^2}{1.064 \times 10^{-6}} \approx \pm 6.184 \text{ mm}.$$

The waist is located approximately 6.2 mm *behind* the collimator face.⁶

3. Recalculating the Rayleigh range with the corrected waist:

$$z_{01} = \frac{\pi \omega_{01}^2}{\lambda} = \frac{\pi \cdot (2.799 \times 10^{-4})^2}{1.064 \times 10^{-6}} \approx 23.132 \text{ cm}.$$

4. Because the beam waist $\omega_{01} \approx 280 \mu\text{m}$ is the *minimum* radius of the laser, the beam can only diverge from that point and would “scrape” the inside of the PPLN crystal. A focusing lens is therefore required.

3.4 Lens Selection

1. The magnification relation from [?]:

$$\omega_{02} = \alpha \cdot \omega_{01} \Rightarrow \alpha = \frac{\omega_{02}}{\omega_{01}}. \quad (6)$$

2. The lens-placement formula from [?]:

$$\alpha = \frac{f}{\sqrt{(|d_1| - f)^2 + z_{01}^2}}. \quad (7)$$

Solving for $|d_1|$:

$$\alpha^2 \left[(|d_1| - f)^2 + z_{01}^2 \right] = f^2 \Rightarrow |d_1| = \sqrt{\left(\frac{f}{\alpha} \right)^2 - z_{01}^2} + f.$$

3. Image-distance formula from [?]:

$$d_2 = f + \alpha^2(d_1 - f). \quad (8)$$

4. **Numerical values.** With $\omega_{02} = 80 \mu\text{m}$ and $\omega_{01} = 279.9 \mu\text{m}$:

$$\alpha = \frac{80}{279.9} \approx 0.28582, \quad z_{01} \approx 0.23132 \text{ m}.$$

For a focal length f chosen in the lab:

$$d_1 = \sqrt{\left(\frac{f}{0.28582}\right)^2 - (0.23132)^2} + f, \quad d_2 = f + (0.28582)^2(d_1 - f).$$

5. **Derivation of equations used above** Starting from [?]:

$$\frac{1}{q_1} = \frac{1}{R_g} - i \frac{\lambda}{\pi \omega_{01}^2}. \quad (9)$$

where R_g is the radius of curvature of the input beam. Taking the limit $R_g \rightarrow \infty$ (collimated input):

$$q_1 = i \frac{\pi \omega_{01}^2}{\lambda} = iz_{01}.$$

The ABCD ray-transfer matrix for free-propagation d_1 , thin lens f , then free-propagation d_2 is:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & d_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix} \begin{pmatrix} 1 & d_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{d_2}{f} & d_1 + d_2 - \frac{d_1 d_2}{f} \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{pmatrix}.$$

so $A = 1 - \frac{d_2}{f}$, $B = d_1 + d_2 - \frac{d_1 d_2}{f}$, $C = -\frac{1}{f}$, $D = 1 - \frac{d_1}{f}$.

The q -parameter transforms as (Eq. (??) from [?]):

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D}. \quad (10)$$

Substituting $q_1 = iz_{01}$:

$$q_2 = \frac{\left(1 - \frac{d_2}{f}\right)iz_{01} + \left(d_1 + d_2 - \frac{d_1 d_2}{f}\right)}{\left(-\frac{1}{f}\right)iz_{01} + \left(1 - \frac{d_1}{f}\right)}.$$

Multiplying numerator and denominator by the complex conjugate of the denominator, $\left(1 - \frac{d_1}{f}\right) + i\left(\frac{z_{01}}{f}\right)$:

Real part:

$$\begin{aligned} \text{Re}(q_2) &= \frac{-\left(1 - \frac{d_2}{f}\right)\left(\frac{z_{01}^2}{f}\right) + \left(d_1 + d_2 - \frac{d_1 d_2}{f}\right)\left(1 - \frac{d_1}{f}\right)}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \\ &= \frac{d_2 \left[\left(1 - \frac{d_1}{f}\right)^2 + \frac{z_{01}^2}{f^2} \right] + d_1 \left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \end{aligned}$$

$$= d_2 + \frac{d_1 \left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2}.$$

Imaginary part:

$$\text{Im}(q_2) = \frac{z_{01} \left[\left(1 - \frac{d_1}{f} - \frac{d_2}{f} + \frac{d_1 d_2}{f^2}\right) + \left(\frac{d_1}{f} + \frac{d_2}{f} - \frac{d_1 d_2}{f^2}\right) \right]}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} = \frac{z_{01}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2}.$$

Full expression:

$$q_2 = \left[d_2 + \frac{d_1 \left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \right] + i \left[\frac{z_{01}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \right].$$

Setting the real part to zero to satisfy the waist condition: $\lim_{z \rightarrow 0} \frac{1}{R_g(z)} = 0$ results in

$$d_2 = \frac{f \left[(d_1 - f)^2 + z_{01}^2 \right] + f^2 (d_1 - f)}{(d_1 - f)^2 + z_{01}^2} = f + \frac{f^2 (d_1 - f)}{(d_1 - f)^2 + z_{01}^2}$$

$$d_2 = f + \alpha^2 (d_1 - f).$$

proving Eq. (??), where $\alpha^2 = \frac{f^2}{(d_1 - f)^2 + z_{01}^2}$ follows Eq. (??).

4 Use Case

PPLN:⁷ To get the most out of our PPLN crystals, four key aspects must be considered:

1. Crystal Length

(a) No special action required at this stage.

2. Polarization

(a) The input light must be e-polarized, i.e. the polarization must be aligned with the dipole moment of the crystal. This is accomplished by aligning the polarization axis of the light parallel to the thickness of the crystal.[?]

3. Focusing and the Optical Arrangement

(a) For SHG with CW lasers, the Boyd–Kleinman result shows that optimum conversion efficiency is achieved when $L/b = 2.84$.[?]

4. Temperature and Period

(a) The crystal is manufactured with periodic stripes of alternating polarity with a spacing called the poling period. This quasi-phase-matched arrangement allows the harmonic power to build up continuously.

- (b) The effective poling period can be tuned by changing the crystal temperature, which causes thermal expansion or contraction.
- (c) Note: The refractive index of the crystal also changes with temperature.
- (d) The output intensity follows a sinc^2 curve (shown below), defining the temperature acceptance bandwidth.
- (e) Acceptance bandwidth is inversely proportional to crystal length.
- (f) The optimum temperature can be found by heating the crystal to 20°C above the calculated phase-matching temperature, then allowing it to cool while monitoring the generated output power.[?]

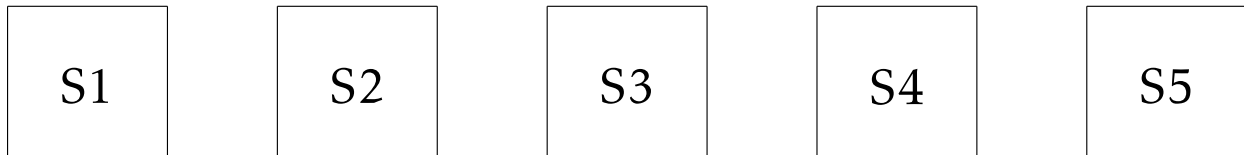


Figure 1: S1: Laser at 1064 nm. S2: [Component — fill in]. S3: [Component — fill in]. S4: [Component — fill in]. S5: [Component — fill in].

Notes

¹Wikipedia entry: Second-harmonic generation (SHG), also known as frequency doubling, is the lowest-order wave-wave nonlinear interaction that occurs in various systems, including optical, radio, atmospheric, and magnetohydrodynamic systems. As a prototype behavior of waves, SHG is widely used, for example, in doubling laser frequencies. SHG was initially discovered as a nonlinear optical process in which two photons with the same frequency interact with a nonlinear material, are “combined”, and generate a new photon with twice the energy of the initial photons (equivalently, twice the frequency and half the wavelength), conserving the coherence of the excitation. It is a special case of two-photon sum-frequency generation, and more generally of harmonic generation.

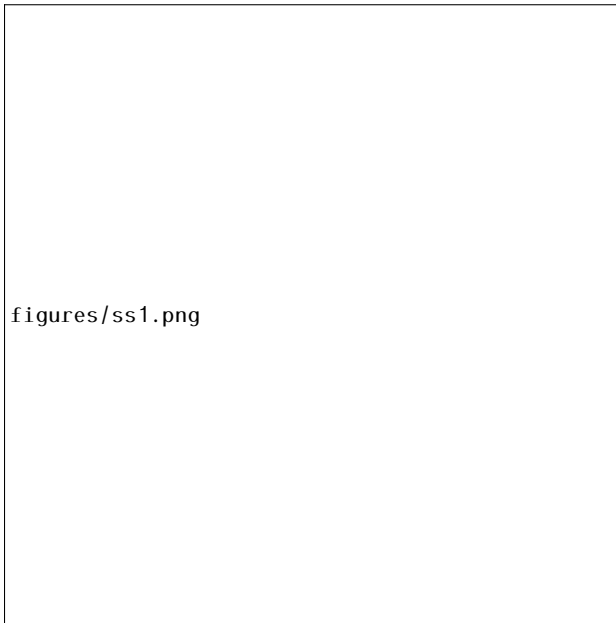
²**Phase-matching:** For sum-frequency generation to occur efficiently, phase-matching conditions must be satisfied: $\hbar k_3 \approx \hbar k_1 + \hbar k_2$, where k_1, k_2, k_3 are the angular wavenumbers of the three waves travelling through the medium. (This equation resembles conservation of momentum.) The sum-frequency generation becomes more efficient as this condition is satisfied more precisely.

³In this calculation I assume the full beam diameter $2\omega_{01} = 0.56$ mm because the beam emerges from a collimator. This assumes the beam waist may be located a short distance away from the collimator face, as noted in [?]: “The purpose of a beam collimator is essentially to transform a strongly diverging light beam into a collimated beam, i.e., a beam where light propagates essentially only in one direction, and the beam divergence is weak. The output beam may have its beam waist close to the output aperture, or a mild focus somewhat away from it.”

⁴“In general, a good rule of thumb is that the spot size should be chosen such that the Rayleigh range is half the length of the crystal.” If $z_{02} = L/2$, then $b = 2z_{02} = L$, so $L/b = 1$.

⁵Note: 1.21 mRad was a measured value, substantially lower than the spec sheet value of approximately 5 mRad.

⁶The following calculations were made under the (incorrect) assumption of a waist radius of $45 \mu\text{m}$ rather than the updated $279.9 \mu\text{m}$ derived here.



⁷Periodically poled lithium niobate (PPLN) is a lithium niobate crystal that switches polarity in a periodic fashion and is used for achieving quasi-phase-matching in nonlinear optics.